Intermediate Macroeconomics

Tutorial 5 Real-Business-Cycle Theory
Answer

1. **Technology shocks and macroeconomic fluctuations in the RBC model.** Consider the RBC model in the lecture notes, with aggregate production

$$Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}, \qquad 0 < \alpha < 1.$$

Let $\widetilde{x}_t = \ln x_t - \ln \overline{x}_t$ denote the deviation of the log of a variable x_t from its balanced-growth-path value.

(a) Technology follows

$$\widetilde{A}_t = \rho_A \widetilde{A}_{t-1} + \varepsilon_{A,t}, \qquad \rho_A = 0.95.$$

The economy is initially on its balanced growth path. At $t = 0$, an unexpected positive innovation of 1 percent occurs ($\varepsilon_{A,0} = 0.01$), and there are no further innovations.

(i) Calculate $\widetilde{A}_0$, $\widetilde{A}_1$, and $\widetilde{A}_2$.

(ii) Write the expression for $\widetilde{A}_t$ for any $t \geq 0$.

(iii) Explain what the persistence parameter ρ_A means.

Answer:

The one-time innovation is $\varepsilon_{A,0} = 0.01$, and $\widetilde{A}_{-1} = 0$. Therefore,

$$\widetilde{A}_0 = 0.01,$$

$$\widetilde{A}_1 = 0.95(0.01) = 0.0095,$$

and

$$\widetilde{A}_2 = 0.95(0.0095) = 0.009025.$$

More generally, in the absence of further innovations,

$$\widetilde{A}_t = 0.01(0.95)^t, \qquad t \geq 0.$$

Thus, the technology deviations in periods 0, 1, and 2 are 1 percent, 0.95 percent, and 0.9025 percent, respectively. The persistence parameter $\rho_A = 0.95$ means that 95 percent of the previous period's deviation remains in the current period. The shock therefore fades gradually rather than disappearing immediately.

(b) Suppose that the calibrated RBC model gives the following log-linearized policy and transition rules:

$$\begin{aligned}
\widetilde{C}_t &= 0.59\widetilde{K}_t + 0.38\widetilde{A}_t - 0.13\widetilde{G}_t, \\
\widetilde{L}_t &= -0.31\widetilde{K}_t + 0.35\widetilde{A}_t + 0.15\widetilde{G}_t, \\
\widetilde{K}_{t+1} &= 0.95\widetilde{K}_t + 0.08\widetilde{A}_t - 0.004\widetilde{G}_t.
\end{aligned}$$

Assume $\alpha = 1/3$, $\widetilde{K}_0 = 0$, and $\widetilde{G}_t = 0$ in all periods. Use the technology shock in part (a).

(i) Starting from the Cobb–Douglas production function, derive the log-linearized expression for $\widetilde{Y}_t$.

(ii) Calculate the percentage responses of consumption, labor, and output at $t = 0$, and the response of capital at $t = 1$.

(iii) Continue the calculations for $t = 1$ and $t = 2$. Complete the table, reporting percentage deviations from the balanced growth path.

Answer:

Taking logs of the Cobb-Douglas production function,

$$Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha},$$

and subtracting the balanced-growth-path expression gives

$$\widetilde{Y}_t = \alpha \widetilde{K}_t + (1-\alpha)(\widetilde{A}_t + \widetilde{L}_t).$$

With $\alpha = 1/3$,

$$\widetilde{Y}_t = \frac{1}{3}\widetilde{K}_t + \frac{2}{3}(\widetilde{A}_t + \widetilde{L}_t).$$

At $t = 0$, $\widetilde{K}_0 = 0$ and $\widetilde{A}_0 = 0.01$. The policy rules imply

$$\widetilde{C}_0 = 0.59(0) + 0.38(0.01) = 0.0038 = 0.38\%,$$

$$\widetilde{L}_0 = -0.31(0) + 0.35(0.01) = 0.0035 = 0.35\%.$$

Therefore,

$$\widetilde{Y}_0 = \frac{1}{3}(0) + \frac{2}{3}(0.01 + 0.0035) = 0.009 = 0.90\%.$$

The capital transition equation gives

$$\widetilde{K}_1 = 0.95(0) + 0.08(0.01) = 0.0008 = 0.08\%.$$

At $t = 1$, $\widetilde{A}_1 = 0.0095$ and $\widetilde{K}_1 = 0.0008$. Hence,

$$\widetilde{C}_1 = 0.59(0.0008) + 0.38(0.0095) = 0.004082 = 0.4082\%,$$

$$\widetilde{L}_1 = -0.31(0.0008) + 0.35(0.0095) = 0.003077 = 0.3077\%,$$

$$\widetilde{Y}_1 = \frac{1}{3}(0.0008) + \frac{2}{3}(0.0095 + 0.003077) \approx 0.0086513 = 0.8651\%.$$

The capital transition equation gives

$$\widetilde{K}_2 = 0.95(0.0008) + 0.08(0.0095) = 0.00152 = 0.152\%.$$

At $t = 2$, $\widetilde{A}_2 = 0.009025$ and $\widetilde{K}_2 = 0.00152$. Thus,

$$\widetilde{C}_2 = 0.59(0.00152) + 0.38(0.009025) = 0.0043263 = 0.43263\%,$$

$$\widetilde{L}_2 = -0.31(0.00152) + 0.35(0.009025) = 0.00268755 = 0.268755\%,$$

$$\widetilde{Y}_2 = \frac{1}{3}(0.00152) + \frac{2}{3}(0.009025 + 0.00268755) \approx 0.0083150 = 0.8315\%.$$

The results are summarized below. All entries in the table are percentage deviations from the balanced-growth path, not growth rates between consecutive periods.

Variable	$t = 0$	$t = 1$	$t = 2$
Technology, $\widetilde{A}_t$	1.0000%	0.9500%	0.9025%
Capital, $\widetilde{K}_t$	0.0000%	0.0800%	0.1520%
Consumption, $\widetilde{C}_t$	0.3800%	0.4082%	0.4326%
Labor, $\widetilde{L}_t$	0.3500%	0.3077%	0.2688%
Output, $\widetilde{Y}_t$	0.9000%	0.8651%	0.8315%

(c) Using your results in part (b), briefly explain:

(i) Why does the capital stock respond with a lag?

(ii) Why can consumption continue to rise even as the technology shock fades?

(iii) What are the two mechanisms through which the initial technology innovation generates persistent macroeconomic fluctuations?

Answer:

Capital is a predetermined state variable. It cannot jump when the shock arrives; it changes through investment and capital accumulation, so the capital response appears in subsequent periods.

Consumption can continue to rise in this calibration because its policy rule depends positively on both technology and capital. Although the technology deviation declines from period to period, the capital stock accumulates, and its positive contribution to consumption more than offsets the fading technology contribution over the first few periods.

The two propagation mechanisms are (1) persistence in the AR(1) technology process and (2) gradual endogenous capital accumulation. The first makes the technology deviation fade slowly, while the second carries the effects of the shock into future periods.

2. **A simplified real-business-cycle model with additive technology shocks.** (Blanchard and Fischer, 1989.) Consider an economy consisting of a constant population of infinitely lived individuals. The representative individual maximizes the expected value of $\sum_{t=0}^{\infty} u\left(C_t\right) / \left(1+\rho\right)^t$, $\rho > 0$. The instantaneous utility function $u\left(C_t\right)$, is $u\left(C_t\right) = C_t - \theta C_t^2$. Assume that C is always in the range where $u'\left(C_t\right)$ is positive.

Output is linear in capital, plus an additive disturbance: $Y_t = AK_t + e_t$. There is no depreciation; thus $K_{t+1} = K_t + Y_t - C_t$, and the interest rate is A. Assume $A = \rho$. Finally, the disturbance follows a first-order autoregressive process: $e_t = \phi e_{t-1} + \varepsilon_t$, where $-1 < \phi < 1$ and where the ε_t's are mean-zero, i.i.d. shocks.

(a) Find the first-order condition (Euler equation) relating C_t and expectations of C_{t+1}.

Answer:

The budget constraint and production function implies $C_t = (1+A)\,K_t - K_{t+1} + e_t$.

Let $V(K_t)$ be the optimal value function of the agent with K_t. The Bellman equation is

$$V(K_t) = \max_{K_{t+1}} \left[(1+A)\,K_t - K_{t+1} + e_t\right] - \theta\left[(1+A)\,K_t - K_{t+1} + e_t\right]^2 + \frac{1}{1+\rho} E_t V(K_{t+1}).$$

FOC:

$$-1 + 2\theta \left[(1 + A) K_t + e_t - K_{t+1} \right] + \frac{1}{1+\rho} E_t V'(K_{t+1}) = 0,$$

after rearrangement,

$$E_t V'(K_{t+1}) = (1 + \rho) (1 - 2\theta C_t) . \tag{1}$$

By the envelope theorem,

$$V'(K_t) = (1 + A) \left\{ 1 - 2\theta \left[(1 + A) K_t + e_t - K_{t+1} \right] \right\}, \tag{2}$$

we have

$$\begin{aligned} V'(K_{t+1}) &= (1 + A) \left\{ 1 - 2\theta \left[(1 + A) K_{t+1} + e_{t+1} - K_{t+2} \right] \right\} \\ &= (1 + A) (1 - 2\theta C_{t+1}) . \end{aligned} \tag{3}$$

Combining (1) and (3) we obtain the intertemporal Euler equation:

$$1 - 2\theta E \left[C_{t+1} \right] = \frac{1+\rho}{1+A} (1 - 2\theta C_t) . \tag{4}$$

Using the assumption that $A = \rho$ and simplifying yields

$$E \left[C_{t+1} \right] = C_t. \tag{5}$$

This equation implies that consumption follows a random walk. The expected value of consumption next period is simply equal to today's actual realization of consumption.

(b) Guess that consumption takes the form $C_t = \alpha + \beta K_t + \gamma e_t$. Given this guess, what is K_{t+1} as a function of K_t and e_t?

Answer:

We will guess that consumption takes the form

$$C_t = \alpha + \beta K_t + \gamma e_t. \tag{6}$$

Substitute equation (6) and the production function, $Y_t = AK_t + e_t$, into the capital accumulation equation, $K_{t+1} = K_t + Y_t - C_t$, to obtain

$$K_{t+1} = K_t + AK_t + e_t - \alpha - \beta K_t - \gamma e_t,$$

or simply,

$$K_{t+1} = -\alpha + (1 + A - \beta) K_t + (1 - \gamma) e_t. \tag{7}$$

(c) What values must the parameters α, β, and γ have for the first-order condition in part (a) to be satisfied for all values of K_t and e_t?

Answer:

Substituting equation (6) and its one-period forward lag into the first-order condition, equation (6):

$$E \left[\alpha + \beta K_{t+1} + \gamma e_{t+1} \right] = \alpha + \beta K_t + \gamma e_t. \tag{8}$$

Substituting equation (7) into above equation yields

$$\alpha + \beta E_t \left[-\alpha + (1 + A - \beta) K_t + (1 - \gamma) e_t \right] + \gamma E_t \left[e_{t+1} \right] = \alpha + \beta K_t + \gamma e_t. \tag{9}$$

Noting that $E_t \left[e_{t+1} \right] = E_t \left[\phi e_t + \varepsilon_t \right] = \phi e_t$, we can collect terms to obtain

$$\alpha (1 - \beta) + \beta (1 + A - \beta) K_t + \left[\beta + \gamma (\phi - \beta) \right] e_t = \alpha + \beta K_t + \gamma e_t. \tag{10}$$

In order for equation (10) to hold, we need the coefficients on K_t and e_t, as well as the constant term, to be the same on both sides. That is,

$$\begin{cases} \beta (1 + A - \beta) = \beta \\ \beta + \gamma (\phi - \beta) = \gamma \\ \alpha (1 - \beta) = \alpha \end{cases} \tag{11}$$

Solving the equation system, we have

$$\beta = A, \ \gamma = \frac{A}{1 - \phi + A}, \ \alpha = 0. \tag{12}$$

Note that we are also ignoring the case in which $\beta = 0$ and $\gamma = 0$ with no restriction on α, as it makes less economic sense.

(d) We assume that ε, and thus e, both equal 0 until some period t. In period t, there is a one-time, positive realization of $\varepsilon_t = 1 - \phi + A$. From period $t + 1$ forward, $\varepsilon = 0$ again. What are the effects of this one-time shock to ε on the paths of Y, K, and C? [*Hint: find changes in Y, K, and C for periods $t = 1, 2, ...n$.*]

Answer:

Substituting parameter solutions (12) into the guess for consumption, (6), and the capital-accumulation equation, equation (7), yields

$$C_t = AK_t + \frac{A}{1 - \phi + A} e_t, \tag{13}$$

and

$$K_{t+1} = K_t + \frac{1 - \phi}{1 - \phi + A} e_t. \tag{14}$$

From equation (14), we can see that K_t is determined by last period's capital stock and last period's realization of e. Therefore, in period t, K_t is unaffected. From the production function, $Y_t = AK_t + e_t$, we have

$$\Delta Y_t = A \Delta K_t + \Delta e_t = 0 + (1 - \phi + A). \tag{15}$$

Thus output in the period of the shock is higher by $(1 - \phi + A)$. From equation (13), the change in consumption is given by

$$\Delta C_t = A \Delta K_t + \left(\frac{A}{1 - \phi + A} \right) \Delta e_t = 0 + \left(\frac{A}{1 - \phi + A} \right) (1 - \phi + A) = A. \tag{16}$$

Thus consumption in the period of the shock is higher by A.

In period $t+1$, even though ε_{t+1} is assumed to be 0 again, e_{t+1} is different than it would have been in the absence of the one-time shock due to the autoregressive form of the e terms. More precisely

$$\Delta e_{t+1} = \phi \Delta e_t = \phi \left(1 - \phi + A\right). \tag{17}$$

Intuitively, last period, output rose by $(1 - \phi + A)$ but consumption rose only by A. The rest of the increase in output, $(1 - \phi)$, was devoted to investment and hence the rise in this period's capital stock by an equal amount (we are assuming no depreciation). From the production function, $Y_{t+1} = AK_{t+1} + e_{t+1}$, the change in output is

$$\Delta Y_{t+1} = A\Delta K_{t+1} + \Delta e_{t+1} = A\left(1 - \phi\right) + \phi\left(1 - \phi + A\right) = A + \phi\left(1 - \phi\right). \tag{18}$$

From equation (13), the change in consumption is given by

$$\begin{aligned} \Delta C_{t+1} &= A\Delta K_{t+1} + \left(\frac{A}{1 - \phi + A}\right)\Delta e_{t+1} \\ &= A\left(1 - \phi\right) + \left(\frac{A}{1 - \phi + A}\right)\phi\left(1 - \phi + A\right) \\ &= A. \end{aligned} \tag{19}$$

Thus there are no further dynamics for consumption. It remains higher than it would have been in the absence of the shock by the amount A.

Similarly, we can calculate these changes for period $t+2$:

$$\Delta e_{t+2} = \phi \Delta e_{t+1} = \phi^2\left(1 - \phi + A\right), \tag{20}$$

$$\Delta K_{t+2} = \Delta K_{t+1} + \frac{1 - \phi}{1 - \phi + A}\Delta e_{t+1} = \left(1 - \phi\right) + \frac{\left(1 - \phi\right)\phi\left(1 - \phi + A\right)}{1 - \phi + A} = 1 - \phi^2, \tag{21}$$

$$\Delta Y_{t+2} = A\Delta K_{t+2} + \Delta e_{t+2} = A\left(1 - \phi^2\right) + \phi^2\left(1 - \phi + A\right) = A + \phi^2\left(1 - \phi\right), \tag{22}$$

$$\Delta C_{t+2} = A\Delta K_{t+2} + \frac{A}{1 - \phi + A}\Delta e_{t+2} = A\left(1 - \phi^2\right) + \frac{A\left(1 - \phi + A\right)}{1 - \phi + A}\phi^2 = A. \tag{23}$$

The pattern can now be inferred. Suppose there is a one-time shock of $\varepsilon_t = 1 - \phi + A$. In the period of the shock, consumption rises by A and permanently stays at that new level with no further dynamics. In addition, n periods after the shock, the change in output is

$$\Delta Y_{t+n} = A + \phi^n\left(1 - \phi\right),$$

and the change in the capital stock is

$$\Delta K_{t+n} = 1 - \phi^n.$$

The nature of the dynamics of Y and K depends on the value of ϕ. In the special case in which ϕ is equal to 0, so that there is no persistence in the technology shock, there are no further dynamics after period $t+1$. The period after the shock, and in all those thereafter, capital is higher by one and output is higher by A.

For the case of $0 < \phi < 1$, the capital stock rises by $(1 - \phi)$ the period after the shock. It

then increases more each period until it asymptotically approaches its new long-run level that is *one* higher than it would have been in the absence of the shock. Output rises by $(1 - \phi + A)$ the period of the shock. It then decreases each period until it asymptotically approaches its new long-run level that is A higher than it would have been in the absence of the shock.

For the case of $-1 < \phi < 0$, capital and output oscillate – alternating above and below their new long-run levels in successive periods – and gradually settle down to be *one* and A higher, respectively.